

EE 506: Solar Photovoltaic Energy Systems

Course Code : EE 506

Course Name : Solar Photovoltaic Energy Systems

L-T-P-C : 2.5-0.5-0-3

Prerequisites : EE309 Power Electronics (or Instructors permission)

Students intended for : B.Tech./M.S./Ph.D

Elective or Compulsory : Elective

Approval: 5th Senate

Course Contents

- **Introduction:** Fossil fuel energy usage and global warming; role of renewable energy in sustainable development; renewable energy sources; global potential for solar electrical energy systems. [3 Lectures]
- **Solar radiation:** Extra terrestrial and terrestrial solar spectrum; clear sky direct-beam radiation; total clear sky insulation on a collecting surface; radiation on the collector in tracking systems; calculation of average monthly insolation from measured data. [9 Lectures]
- **PV cells and modules:** Photovoltaic cell and its simple model; i-v and p-v characteristics; PV modules and arrays; effect of shading, use of bypass and blocking diodes; influence of temperature; types of solar cells and their performance; schemes for maximum power point tracking; solar PV concentrators. [6 Lectures]
- **PV inverters:** Grid-connected single phase PV inverter schemes and control; power processing schemes based on single string, multi-string and ac module technologies; types of grid interface; power electronic converters used in single phase PV systems and their operation; transformer less inverters, centralized grid-connected three- phase inverters for large PV installations. [9 Lectures]
- **Schemes with battery energy storage:** Power processing schemes and control for stand-alone applications; batteries for energy storage types, charging, battery sizing and turn-around efficiency; other types of energy storage for PV systems; grid connected schemes with standby energy storage. [9 Lectures]
- **System level issues:** Design related issues; grounding, dc arcing and other safety related issues; islanding; harmonics; electro- magnetic interference; energy yield and economics of a PV installation. [6 Lectures]

Text Book:

1. Gilbert M. Masters, **Renewable and Efficient Electric Power Systems**, John Wiley & Sons, 2004

Reference Books:

1. Roger A. Messenger & Jerry Ventre, **Photovoltaic Systems Engineering**, 2nd Edition, CRC Press, 2004.
2. Solanki, **Solar Photovoltaics: Fundamentals, Technologies and Applications**, PHI Learning Pvt Ltd, 2009.